Analysis of ^{48}Ti

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0.1 Energy and Efficiency Callibration

Table 1: Efficiency calibration data of the Clover detectors using standard radioactive sources ^{152}Eu and ^{66}Ga over a wide energy range.

Energy of gamma(E_γ)	Efficiency
121.78	3.18
244.69	2.31
344.38	1.89
411.12	1.67
443.96	1.61
778.91	1.12
867.38	1.06
964.08	1.00
1085.87	0.92
1212.95	0.85
1299.14	0.81
1408.01	0.77
1918.34	0.53
2422.54	0.45
3228.82	0.35
3380.88	0.32
3791.04	0.26
4085.88	0.24
4461.25	0.23
4806.06	0.22

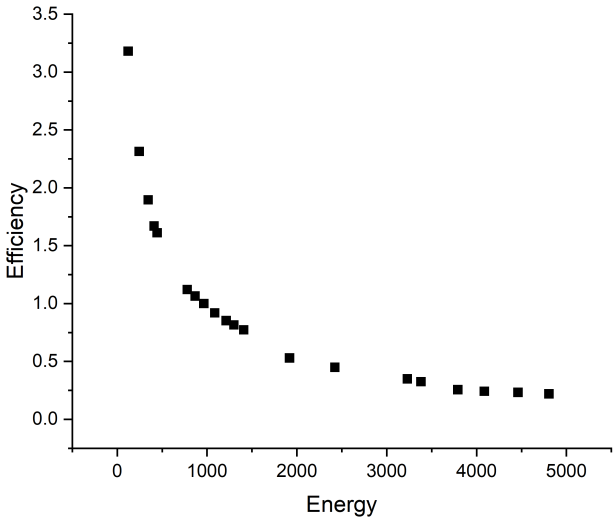

Figure 1: Efficiency calibration curve of the Clover detectors using standard radioactive sources ^{152}Eu and ^{66}Ga.

Table 2: Measured Relative intensity I_{rel}, DCO ratios, and IPDCO values for various γ-ray transitions in ^{48}Ti.

Transition $E_i \rightarrow E_f$ (keV)	$J_i^\pi \rightarrow J_f^\pi$	E_γ (keV)	I_{rel}	Gated γ (keV)	RDCO	IPDCO
983 → 0	$2^+ \rightarrow 0^+$	983	100.0(60)	1037	0.99(7)	0.18(1)
2295 → 983	$4^+ \rightarrow 2^+$	1312	69.4(40)	1037	0.98(7)	0.08(1)
2420 → 983	$2^+ \rightarrow 2^+$	1437	6.0(3)	983	0.87(10)	0.05(4)
3223 → 983	$3^+ \rightarrow 2^+$	2240	2.4(3)	983	1.08(20)	0.22(8)
3223 → 2295	$3^+ \rightarrow 4^+$	928	0.9(1)	1312	0.48(17)	-0.09(6)
3239 → 2295	$4^+ \rightarrow 4^+$	944	7.6(4)	1312	1.00(11)	0.06(3)
3332 → 2295	$6^+ \rightarrow 4^+$	1037	29.3(20)	983	1.00(7)	0.03(1)
3358 → 983	$3^- \rightarrow 2^+$	2375	9.7(8)	983	1.02(11)	-0.05(3)
3371 → 983	$2^+ \rightarrow 2^+$	2388	1.2(1)	983	0.92(22)	0.05(1)
3507 → 2295	$6^+ \rightarrow 4^+$	1212	2.5(3)	1312	0.97(14)	0.12(6)
3507 → 3332	$6^+ \rightarrow 6^+$	175	5.2(2)	1037	1.07(8)	0.16(1)
3781 → 3358	$4^- \rightarrow 4^-$	423	2.0(1)	983	0.99(12)	-0.09(4)
3781 → 2295	$4^- \rightarrow 4^+$	1486	1.4(1)	983	0.87(21)	-0.12(8)
3851 → 983	$3^- \rightarrow 2^+$	2868	2.8(4)	983	0.98(18)	-0.13(8)
4045 → 2295	$5^- \rightarrow 4^+$	1750	3.6(4)	1312	0.92(12)	-0.10(4)
4563 → 3332	$8^+ \rightarrow 6^+$	1231	11.9(9)	1312	0.97(8)	0.09(2)
4563 → 3507	$8^+ \rightarrow 6^+$	1057	0.5(1)	1312	1.05(19)	0.07(3)
4956 → 3781	$6^- \rightarrow 4^-$	1174	0.7(1)	1312	1.15(32)	0.24(12)
5312 → 3332	$5^- \rightarrow 6^+$	1980	1.0(1)	1037	0.49(8)	0.18(6)
5197 → 4563	$8^+ \rightarrow 8^+$	633	2.4(2)	1037	1.06(12)	0.19(6)
6034 → 5197	$9^+ \rightarrow 8^+$	837	2.3(4)	1037	0.65(10)	-0.18(8)
6034 → 4563	$9^+ \rightarrow 8^+$	1470	2.3(2)	1231	0.58(6)	-0.07(5)
6103 → 4563	$10^+ \rightarrow 8^+$	1539	3.7(3)	983	0.93(11)	0.14(3)
6880 → 5312	$7^- \rightarrow 5^-$	1568	1.3(2)	1312	1.06(20)	0.41(13)
6907 → 6034	$11^+ \rightarrow 10^+$	872	1.4(2)	1231	0.59(8)	-0.24(12)
7375 → 6103	$11^+ \rightarrow 10^+$	1272	1.8(2)	1037	0.54(8)	-0.05(3)
7375 → 6907	$11^+ \rightarrow 10^+$	468	0.3(1)	1231	0.59(10)	-0.07(2)
8092 → 7375	$12^+ \rightarrow 11^+$	717	1.5(1)	1312	0.53(9)	-0.05(3)

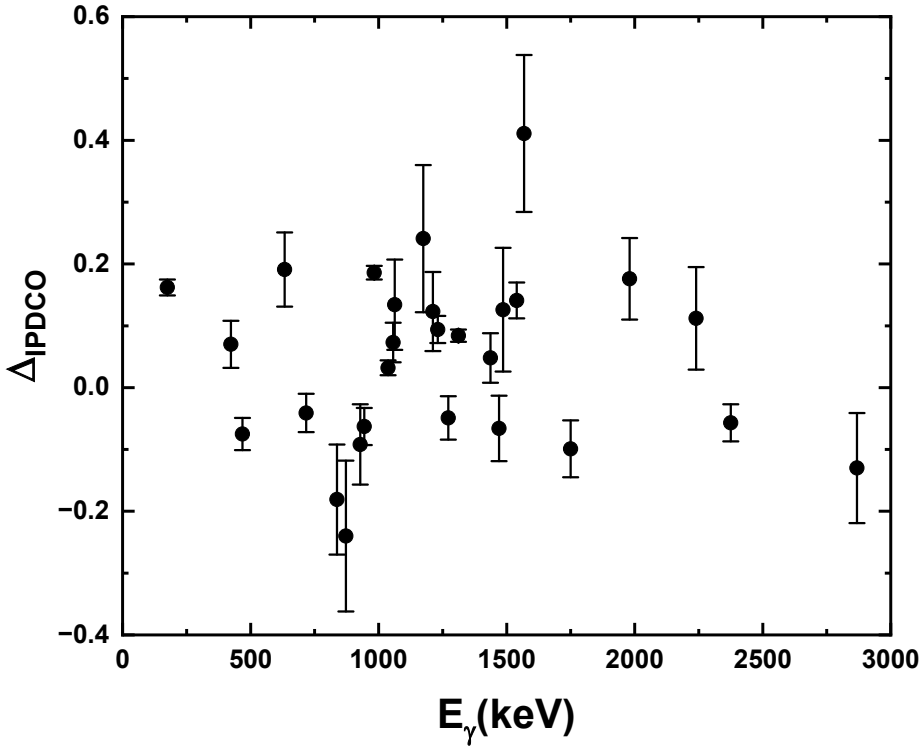

Figure 2: Variation of calculated IPDCO values as a function of γ-ray energy, used to distinguish electric and magnetic character.

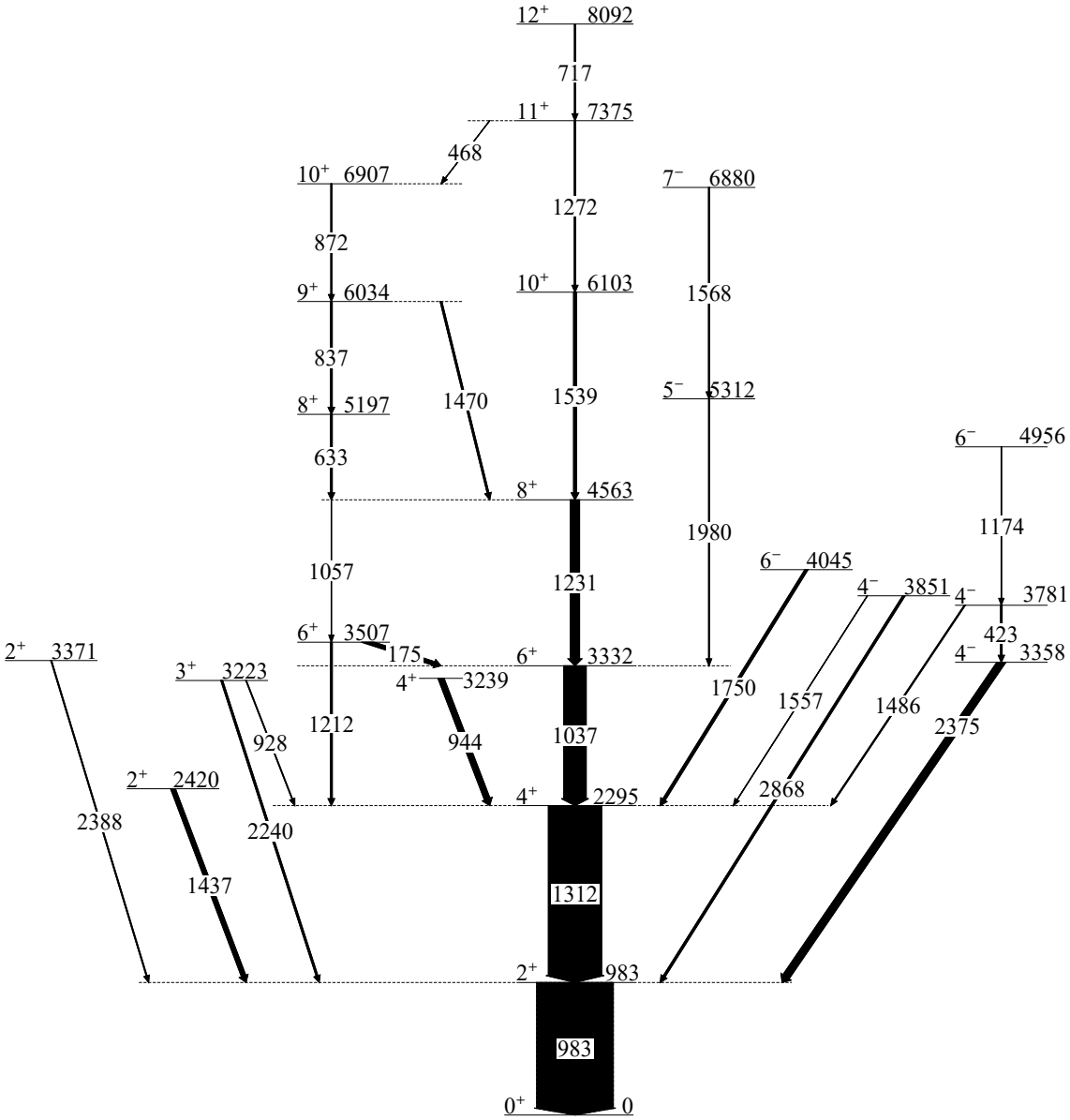

Figure 3: Level scheme of ^{48}Ti as determined from the present spectroscopic study. Transition energies are shown in keV.